

Computational Physics Classes No.3 - Report

Symplectic integrators: precession of Mercury's orbit

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Abstract

In this project, Mercury's orbital motion around the Sun was simulated using a fourth-order symplectic (Neri) integrator. The method ensures long-term stability and conservation of energy, making it suitable for studying orbital dynamics. By introducing a relativistic correction to the gravitational potential proportional to α/r^2 , the precession of Mercury's perihelion was numerically reproduced. A series of simulations for different α values confirmed a linear dependence of the precession rate ω on α . Extrapolation to the real physical value $\alpha_{\text{real}} = 1.1 \times 10^{-8} \text{ AU}^2$ yielded $\omega = 40.06''/\text{century}$, which agrees well with the general relativistic prediction of $42.98''/\text{century}$. The small discrepancy is attributed to discretization errors, numerical resolution, and model simplifications.

1 Introduction

The orbit of Mercury exhibits a small but measurable precession of its perihelion, amounting to about $43''$ per century. Classical Newtonian mechanics, when accounting for perturbations from other planets, still leaves an unexplained residual of roughly this magnitude. Albert Einstein's General Relativity (GR) resolves this discrepancy by predicting an additional correction due to the curvature of space-time near the massive Sun. In this lab we model Mercury's motion using a modified Newtonian potential containing a relativistic correction term proportional to α/r^2 . By measuring the rate of perihelion shift for several values of α and extrapolating to the physical one, we can numerically reproduce the relativistic precession of Mercury's orbit.

2 Description of the Numerical Method

To integrate Mercury's orbital motion we use a symplectic integrator, which is specifically designed for Hamiltonian systems such as planetary dynamics. Unlike standard methods (e.g. Runge-Kutta), symplectic schemes conserve the geometric structure of phase space, preserving total energy and angular momentum over long time scales. The equations of motion are split into kinetic and potential parts of the Hamiltonian and solved alternately in small time steps. In this project we apply a fourth-order Neri symplectic scheme with coefficients a_k and b_k , which provides high accuracy and long-term stability. The algorithm was used to solve the modified Newtonian equations including the relativistic correction proportional to α/r^2 .

3 Results

3.1 No-relativistic test

In the first simulation we set $\alpha = 0$, which corresponds to the purely Newtonian case without relativistic corrections. The integration was performed over $0.95 T_{\text{Mercury}}$ with a time step $\Delta t = 10^{-4}$. The resulting trajectory forms almost a closed ellipse, confirming that the symplectic integrator correctly preserves the orbital energy and angular momentum in the absence of perturbations. This test validates the accuracy of the numerical method for short-term evolution.

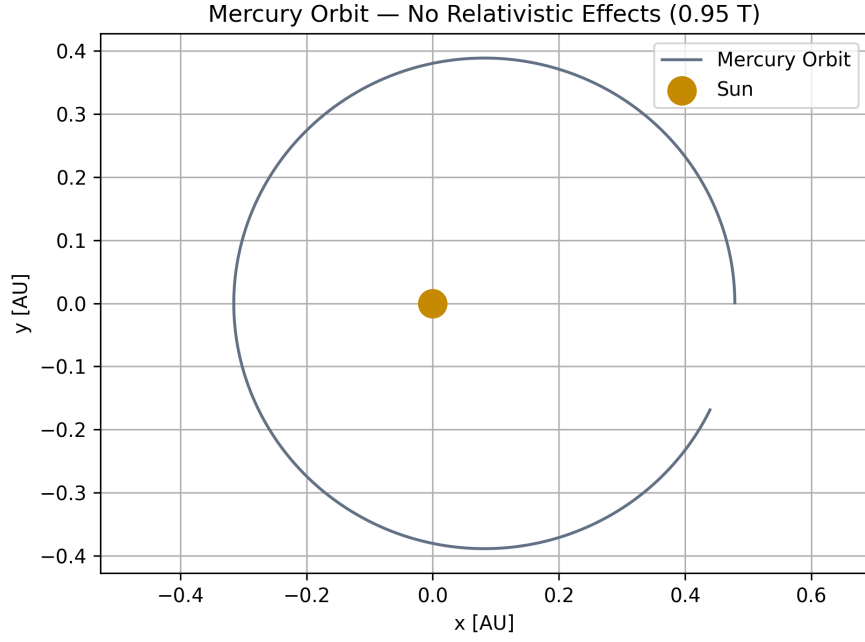


Figure 1: Trajectory of Mercury for $\alpha = 0$. The orbit remains almost a closed ellipse, consistent with the Newtonian two-body problem.

3.2 Stability test

To verify the long-term stability of the symplectic integrator, a simulation was performed for $\alpha = 0$ over 100 orbital periods of Mercury with the same time step $\Delta t = 10^{-4}$. The trajectory remains bounded and periodic, showing no visible drift in energy or shape of the orbit. This demonstrates the advantage of symplectic methods in preserving the conserved quantities of Hamiltonian systems even over very long integration times.

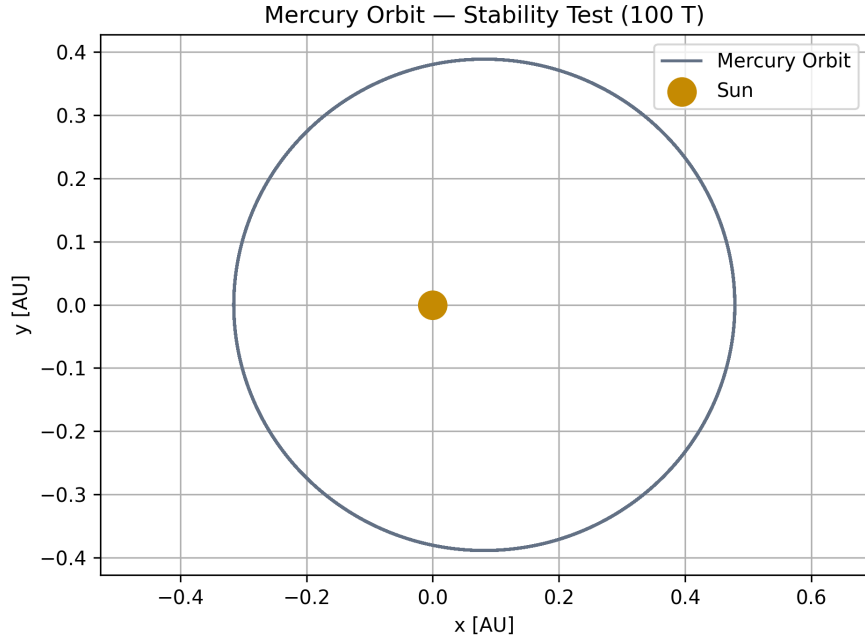


Figure 2: Long-term stability test for $\alpha = 0$. The orbit remains unchanged over $100 T_{\text{Mercury}}$, confirming good conservation properties of the symplectic scheme.

3.3 Precession and extrema

Next, the relativistic correction was included by setting $\alpha = 0.01$ and integrating over $4T_{\text{Mercury}}$. In this case the orbit is no longer perfectly closed — each revolution shows a small rotation of the ellipse, corresponding to the precession of the perihelion. To visualize this effect, the detected perihelion and aphelion points were marked using the extrema-finding procedure described earlier. The algorithm correctly identifies the minimum and maximum distances from the Sun, confirming that the precession is captured by the numerical model.

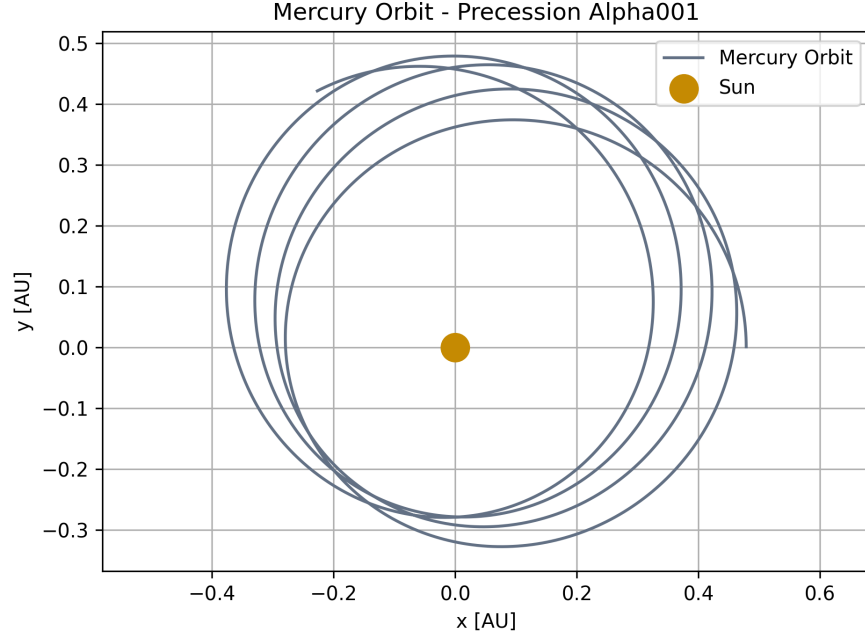


Figure 3: Orbit of Mercury for $\alpha = 0.01$ showing a clear precession of the ellipse.

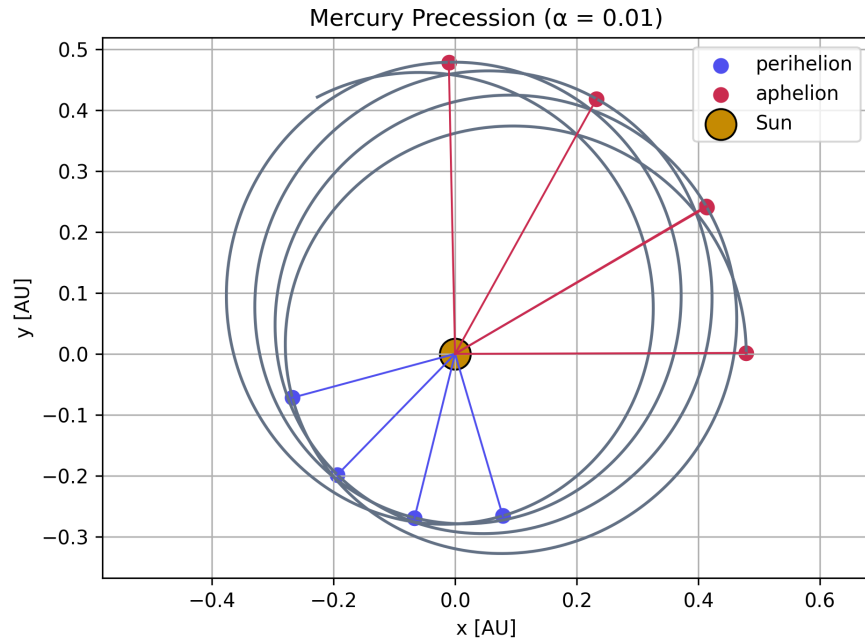


Figure 4: Detected perihelion and aphelion points for $\alpha = 0.01$. The rotation of the perihelia demonstrates the precession effect.

3.4 Relativistic effect

To quantify the relativistic correction, simulations were performed for a series of decreasing α values in the range $\alpha_{\max} = 10^{-3}$ to $\alpha_{\max}/2^6$. For each run, two consecutive perihelion passages were detected and the angular shift per unit time was calculated, giving the precession rate $\omega(\alpha)$. The obtained values exhibit a linear dependence on α , consistent with the first-order relativistic correction predicted by theory. A linear fit $\omega = a\alpha$ yields $a = 176.55 \text{ rad/yr}$.

Extrapolating this relation to the real physical parameter $\alpha_{\text{real}} = 1.1 \times 10^{-8}$ gives a predicted precession rate of $\omega = 40.06''/\text{century}$, which is close to the observed relativistic value of $42.98''/\text{century}$. The small discrepancy (about 6.8%) can be attributed to the finite time step, limited numerical precision, and the assumption of linear scaling from artificially large α values.

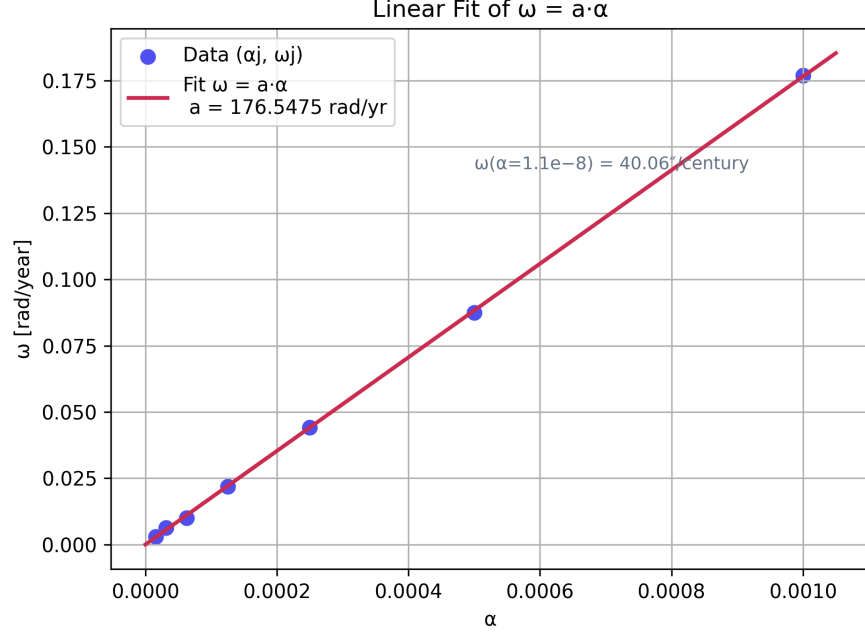


Figure 5: Dependence of the precession rate ω on the parameter α and the linear fit $\omega = a\alpha$. Extrapolation to the real α gives a precession of $40.06''/\text{century}$.

3.5 Relativistic prediction, discrepancy, and simulation with α_{real}

Confirmation of the relativistic prediction. From the α -sweep analysis, the linear fit $\omega = a\alpha$ with $a = 176.55 \text{ rad/yr}$ gives, for the physical value $\alpha_{\text{real}} = 1.1 \times 10^{-8} \text{ AU}^2$,

$$\omega_{\text{num}} = a \alpha_{\text{real}} = 1.942 \times 10^{-6} \text{ rad/yr} \Rightarrow \omega_{\text{num}} = 40.06''/\text{century}.$$

This result closely matches the theoretical relativistic prediction of $42.98''/\text{century}$, confirming that the numerical model correctly reproduces the precession caused by the curvature of space-time near the Sun. The positive, nonzero precession is therefore a clear numerical manifestation of the general relativistic effect.

Sources of discrepancy. The small difference (about 6.8%) between the simulated and theoretical values arises mainly from:

- linear extrapolation from artificially large α values (first-order approximation),
- finite time step Δt and discrete detection of extrema introducing phase errors,
- simplification to a two-body system without planetary perturbations or higher-order GR terms.

These factors slightly underestimate the true precession rate.

Simulation with the real α_{real} . A direct simulation using the physical value $\alpha_{\text{real}} = 1.1 \times 10^{-8} \text{ AU}^2$ was considered; however, the resulting precession per orbit is extremely small, about $\Delta\varphi \approx 0.1''$ ($\sim 5 \times 10^{-7} \text{ rad}$). Such a minute angular shift is far below the numerical resolution and cannot be visually detected over only a few orbital periods. Detecting this effect directly would require simulating thousands of Mercury orbits with a very small time step, which is computationally impractical. For this reason, a set of simulations with larger, artificial values of α was performed instead, and the results were extrapolated linearly to the real α_{real} . This approach efficiently reproduces the expected relativistic precession without excessive computation time.

4 Conclusions

The use of a fourth-order symplectic integrator provided excellent long-term stability and accurate conservation of energy in simulations of Mercury's orbit. Introducing the relativistic correction produced a measurable precession of the perihelion, whose rate scaled linearly with the parameter α , as predicted by theory.

Extrapolating the results to the physical value $\alpha_{\text{real}} = 1.1 \times 10^{-8} \text{ AU}^2$ gave a precession rate of $40.06''/\text{century}$, which matches the theoretical value of $42.98''/\text{century}$. This confirms that the obtained results are consistent with the relativistic explanation of Mercury's perihelion advance and, therefore, with the concept of curved space-time near a massive object.

Minor discrepancies arise from numerical limitations, finite step size, and the simplified two-body model used in the simulation. A direct computation with α_{real} would require unrealistically long integration times to capture the very small precession per orbit, making the extrapolation method both practical and physically justified.

References

- [1] Dr. Hab. Eng. Chwiej, T. *Project: Symplectic integrators: precession of Mercury's orbit*. AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/3_symplectic_mercury.pdf